An Abel Half-Wave Trace for Square-Billiard Clean Families

Route-A structural certificate C157

Abstract

We derive a two-dimensional Poisson formula for the genuine Dirichlet Abel half-wave trace of the unit square. Its nonaxis dual terms reorganize exactly by ordered positive primitive billiard directions and repetitions, retaining clean-family lengths and collisions. Exact shell ledgers and tail-bounded complex sentinels certify the formula. The result is a source trace and natural quantization, not an isolated-orbit determinant or target identity.

Keywords: square billiard; Abel trace; half-wave operator; Poisson summation; clean families; primitive directions.

1 Dirichlet Abel trace

For the unit-square Dirichlet Laplacian, set

$$W_D(s) = \sum_{j,k\geq 1} e^{-\pi s\sqrt{j^2+k^2}}, \qquad \Re s > 0. \tag{1}$$

With the convention $\widehat{f}(m) = \int_{\mathbb{R}^2} f(x)e^{-2\pi i m\cdot x}\,dx$, radial transformation gives

$$\widehat{e^{-\pi s|x|}}(m) = \frac{2s}{\pi(s^2+4|m|^2)^{3/2}}. \tag{2}$$

For real positive s, Poisson summation and $\Theta = 1 + 4/(e^{\pi s}-1) + 4W_D$ yield

$$W_D(s) = \frac{s}{2\pi}\sum_{m\in\mathbb{Z}^2}(s^2+4|m|^2)^{-3/2} - \frac{1}{4} - \frac{1}{e^{\pi s}-1}. \tag{3}$$

The primal exponential sum and dual $|m|^{-3}$ sum converge locally normally. For $\Re s > 0$, $s^2+4|m|^2$ avoids the nonpositive real axis, so the principal power is holomorphic; analytic continuation proves (3) on the half-plane.

2 Primitive directions and repetitions

The dual zero mode is $1/(2\pi s^2)$ and the four axes contribute

$$\frac{2s}{\pi}\sum_{r\geq 1}(s^2+4r^2)^{-3/2}. \tag{4}$$

Every nonaxis vector is uniquely $(\pm ra, \pm rb)$ with $a, b \geq 1$, $\gcd(a,b) = 1$. Thus the interior is

$$\frac{2s}{\pi}\sum_{\substack{a,b\geq 1\\(a,b)=1}}\sum_{r\geq 1}\left(s^2+r^2L_{a,b}^2\right)^{-3/2}, \qquad L_{a,b} = 2\sqrt{a^2+b^2}. \tag{5}$$

Four sign lifts produce the coefficient in (5), while ordered coordinate swaps remain distinct. For $s = \epsilon - it$, its principal-power branch points occur at $t = \pm rL_{a,b}$; these are the square-billiard clean-family lengths with every repetition retained.

3 Certificate and Route-A boundary

Through squared norm 500, the exact ledger contains 98 primitive shells, 239 ordered primitive directions, 161 occupied dual shells, and 373 ordered positive vectors. The first fourfold primitive collision is

$$(1,8),(4,7),(7,4),(8,1), \qquad a^2+b^2 = 65.$$

Two complex sentinels compare (1) against an Epstein-accelerated form of (3), with absolute differences 5.18×10^{-13} and 3.93×10^{-12} inside explicit tail bounds.

The tuple is (`A1_WEAK`, `A2_FAIL`, `A3_FAIL`, `A4_NATURAL_QUANTIZATION`). Although $W_D = \operatorname{Tr} e^{-s\sqrt{\Delta_D}}$ is a genuine source trace, the periodic sets are clean families. We claim no isolated-orbit determinant or stability amplitude, target trace/divisor/functional equation or counting law, arithmetic local/Euler factor, root number, automorphy, Hilbert–Pólya construction, or Route-B authorization. Scope: `NO_BAD_EULER_OR_ROOT_NUMBER`.